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Probability and Statistics Lab (DA-2)

1. Question 1: A transport analyst wants to study the relationship between daily advertising expenditure (X) in thousand rupees and number of passengers (Y) using a new ride-sharing service over 8 days.

Day	Advertising Expenditure (X)	Number of Passengers (Y)
1	5	120
2	8	150
3	6	135
4	10	180
5	7	145
6	12	200
7	9	165
8	11	190

Calculate Karl Pearson's coefficient of correlation between advertising expenditure and number of passengers.

CODE:-

DA1.R

```r
# ------------------------------
# 1. Enter the Data
# ------------------------------

# Advertising Expenditure (in thousand rupees)
X <- c(5, 8, 6, 10, 7, 12, 9, 11)

# Number of Passengers
Y <- c(120, 150, 135, 180, 145, 200, 165, 190)

# ------------------------------
# 2. Descriptive Statistics
# ------------------------------

mean_X <- mean(X)
mean_Y <- mean(Y)

sd_X <- sd(X)
sd_Y <- sd(Y)

cat("Mean of X:", mean_X, "\n")
cat("Mean of Y:", mean_Y, "\n")
cat("Standard Deviation of X:", sd_X, "\n")
cat("Standard Deviation of Y:", sd_Y, "\n\n")

# ------------------------------
# 3. Karl Pearson Correlation
# ------------------------------

correlation <- cor(X, Y, method = "pearson")
cat("Karl Pearson Correlation (r):", correlation, "\n\n")

# ------------------------------
# 4. Correlation Test
# ------------------------------

cor_test <- cor.test(X, Y, method = "pearson")
print(cor_test)

```

80:1 (Untitled) R Script

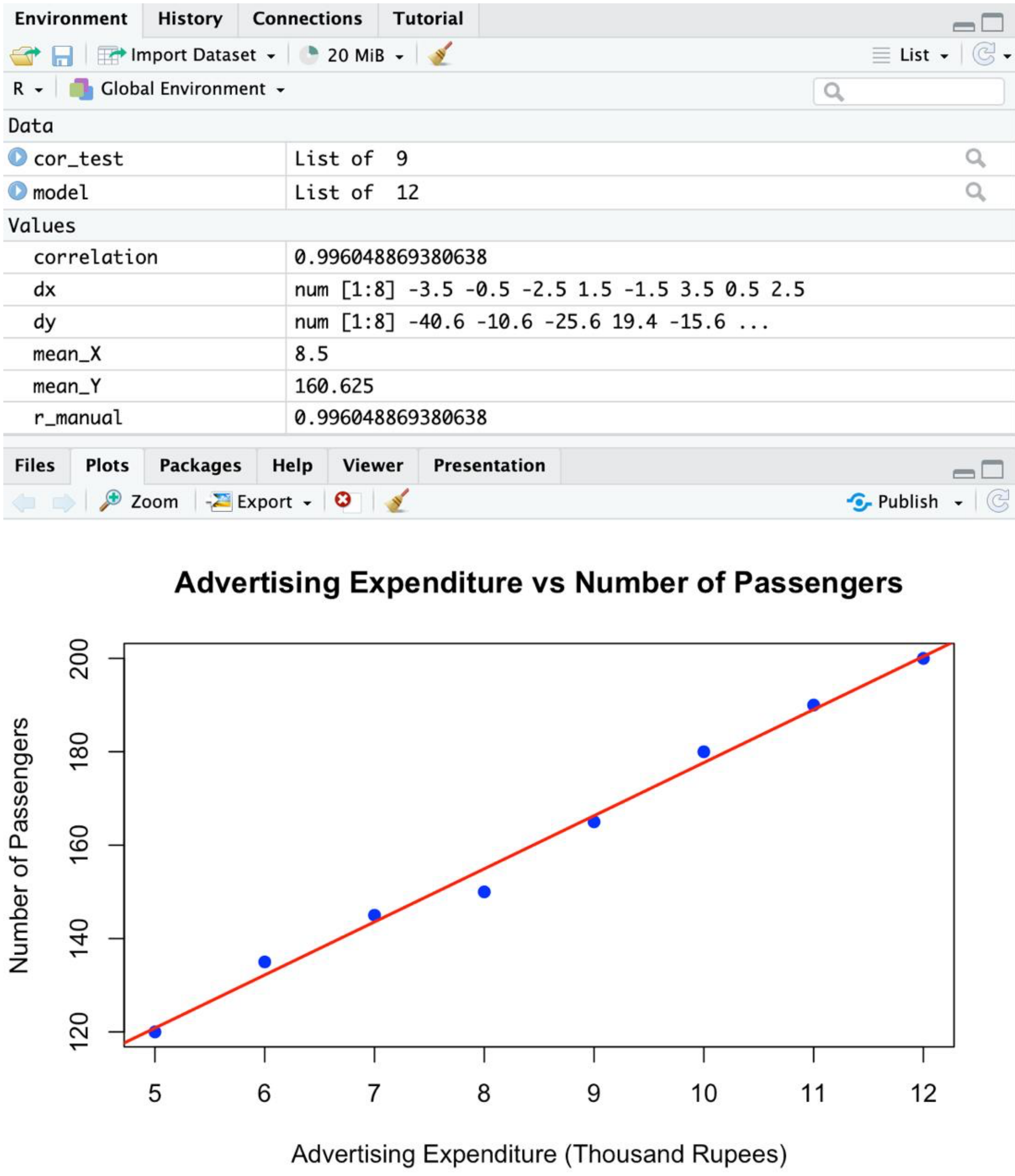
CONSOLE:-

Console Terminal Background Jobs

R 4.5.2 · ~/new da/

```
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> # ------------------------------
> # End of Code
> # ------------------------------
> # ------------------------------
> # End of Code
> # ------------------------------
> # ------------------------------
> # End of Code
> # ------------------------------
> source("~/new da/aryan/DA1.R")
Mean of X: 8.5
Mean of Y: 160.625
Standard Deviation of X: 2.44949
Standard Deviation of Y: 27.95883

Karl Pearson Correlation (r): 0.9960489


        Pearson's product-moment correlation

data:  X and Y
t = 27.473, df = 6, p-value = 1.538e-07
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9774065 0.9993144
sample estimates:
      cor
0.9960489


Manual Calculation of r: 0.9960489

> 
```

OUTPUT:-



Que 1 (code part):-

----

1. Enter the Data

Advertising Expenditure (in thousand rupees)

X <- c(5, 8, 6, 10, 7, 12, 9, 11)

Number of Passengers

Y <- c(120, 150, 135, 180, 145, 200, 165, 190)

----

2. Descriptive Statistics

----

mean_X <- mean(X) mean_Y <- mean(Y)

sd_X <- sd(X) sd_Y <- sd(Y)

cat("Mean of X:", mean_X, "\n") cat("Mean of Y:", mean_Y, "\n") cat("Standard Deviation of X:", sd_X, "\n") cat("Standard Deviation of Y:", sd_Y, "\n\n")

----

3. Karl Pearson Correlation

----

correlation <- cor(X, Y, method = "pearson") cat("Karl Pearson Correlation (r):", correlation, "\n\n")

----

4. Correlation Test

----

cor_test <- cor.test(X, Y, method = "pearson") print(cor_test)

----

5. Manual Calculation of r

Deviations

dx <- X - mean_X dy <- Y - mean_Y

Formula components

sum_dxdy <- sum(dx * dy) sum_dx2 <- sum(dx^2) sum_dy2 <- sum(dy^2)

r_manual <- sum_dxdy / sqrt(sum_dx2 * sum_dy2)

cat("\nManual Calculation of r:", r_manual, "\n\n")

----

6. Simple Linear Regression

----

model <- lm(Y ~ X) summary(model)

----

7. Scatter Plot with Regression Line

----

plot(X, Y, main = "Advertising Expenditure vs Number of Passengers", xlab = "Advertising Expenditure (Thousand Rupees)", ylab = "Number of Passengers", pch = 19, col = "blue")

abline(model, col = "red", lwd = 2)

End of Code

QUE-2

2. A researcher ranked 10 employees based on Performance Score (X) and Leadership Score (Y) as shown below:

Employee	X	Y
1	85	88
2	78	75
3	92	90
4	70	72
5	88	85
6	75	78
7	80	82
8	90	92
9	68	70
10	82	80

Compute Spearman's rank correlation coefficient.

CODE:-

```r
# ------------------------------
# 1. Enter the Data
# ------------------------------

# Performance Score (X)
X <- c(85, 78, 92, 70, 88, 75, 80, 90, 68, 82)

# Leadership Score (Y)
Y <- c(88, 75, 90, 72, 85, 78, 82, 92, 70, 80)

# ------------------------------
# 2. Spearman Rank Correlation
# ------------------------------

spearman_corr <- cor(X, Y, method = "spearman")

cat("Spearman's Rank Correlation (rho):", spearman_corr, "\n\n")

# ------------------------------
# 3. Significance Test
# ------------------------------

spearman_test <- cor.test(X, Y, method = "spearman")

print(spearman_test)

# ------------------------------
# 4. Manual Calculation Method
# ------------------------------

# Rank the data
rank_X <- rank(X)
rank_Y <- rank(Y)

# Difference in ranks
d <- rank_X - rank_Y
d2 <- d^2

# Apply Spearman formula
```

CONSOLE:-

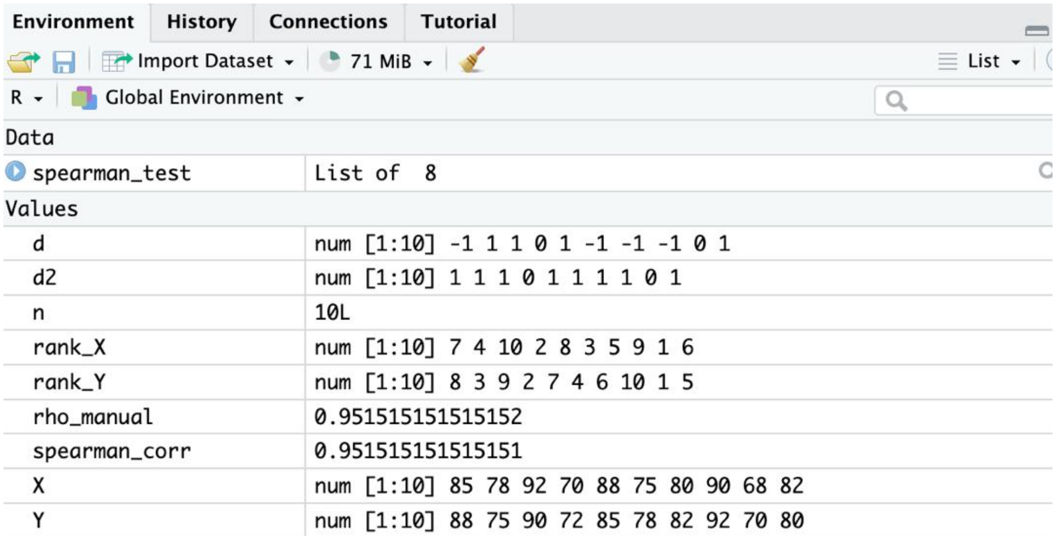
Source

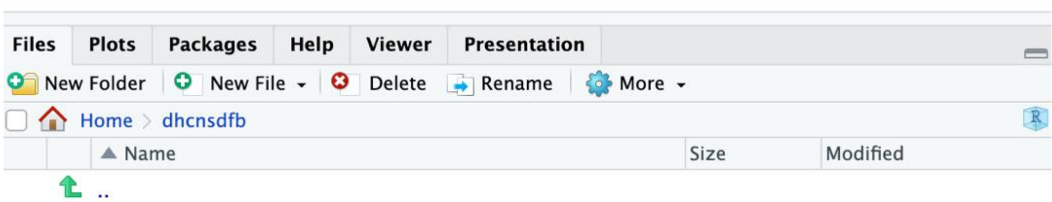
Console | Terminal | Background Jobs

R 4.5.2 · ~/dhcnsdfb/

```
> spearman_test <- cor.test(X, Y, method = "spearman")
>
> print(spearman_test)

        Spearman's rank correlation rho

data:  X and Y
S = 8, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.9515152

>
> # ------------------------------
> # 4. Manual Calculation Method
> # ------------------------------
>
> # Rank the data
> rank_X <- rank(X)
> rank_Y <- rank(Y)
>
> # Difference in ranks
> d <- rank_X - rank_Y
> d2 <- d^2
>
> # Apply Spearman formula
> n <- length(X)
> rho_manual <- 1 - (6 * sum(d2)) / (n * (n^2 - 1))
>
> cat("\nManual Spearman's rho:", rho_manual, "\n")

Manual Spearman's rho: 0.9515152
>
> # ------------------------------
> # End of Code
> # ------------------------------
>
```

OUTPUT:-





Que –2(code part):-

----

1. Enter the Data

Performance Score (X)

```
X <- c(85, 78, 92, 70, 88, 75, 80, 90, 68, 82)
```

Leadership Score (Y)

```
Y <- c(88, 75, 90, 72, 85, 78, 82, 92, 70, 80)
```

----

2. Spearman Rank Correlation

```
spearman_corr <- cor(X, Y, method = "spearman")

cat("Spearman's Rank Correlation (rho):", spearman_corr, "\n\n")
```

---

3. Significance Test

---

spearman_test <- cor.test(X, Y, method = "spearman")

print(spearman_test)

---

4. Manual Calculation Method

Rank the data

rank_X <- rank(X) rank_Y <- rank(Y)

Difference in ranks

d <- rank_X - rank_Y d2 <- d^2

Apply Spearman formula

n <- length(X) rho_manual <- 1 - (6 * sum(d2)) / (n * (n^2 - 1))

cat("\nManual Spearman's rho:", rho_manual, "\n")

End of Code

QUESTION 3:-

3. A logistics company wants to examine how daily fuel consumption depends on the total distance travelled by delivery vehicles.

   The following data were collected for 15 working days:

Day	Distance Travelled (km) (X)	Fuel Consumption (litres) (Y)
1	120	18
2	150	22
3	180	26
4	200	29
5	160	24
6	170	25
7	140	21
8	210	31
9	190	27
10	220	33
11	155	23
12	165	24
13	175	26
14	205	30
15	195	28

   Fit a simple linear regression line of fuel consumption (Y) on distance travelled (X). Estimate the fuel consumption when the distance travelled is 185 km.

4. A retail company wants to analyse how monthly sales revenue depends on advertising

CODE:-

Untitled.R*

```r
# ------------------------------
# 1. Enter the Data
# ------------------------------

# Distance Travelled (km)
X <- c(120,150,180,200,160,170,140,210,190,220,
       155,165,175,205,195)

# Fuel Consumption (litres)
Y <- c(18,22,26,29,24,25,21,31,27,33,
       23,24,26,30,28)

# ------------------------------
# 2. Fit Linear Regression Model
# ------------------------------

model <- lm(Y ~ X)

# Display model summary
summary(model)

# ------------------------------
# 3. Regression Equation
# ------------------------------

intercept <- coef(model)[1]
slope <- coef(model)[2]

cat("Regression Equation:\n")
cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")

# ------------------------------
# 4. Prediction for X = 185 km
# ------------------------------

predicted_value <- predict(model, newdata = data.frame(X = 185))

cat("Estimated fuel consumption for 185 km:",
    round(predicted_value,4), "litres\n")
```

CONSOLE:-

Console | Terminal | Background Jobs

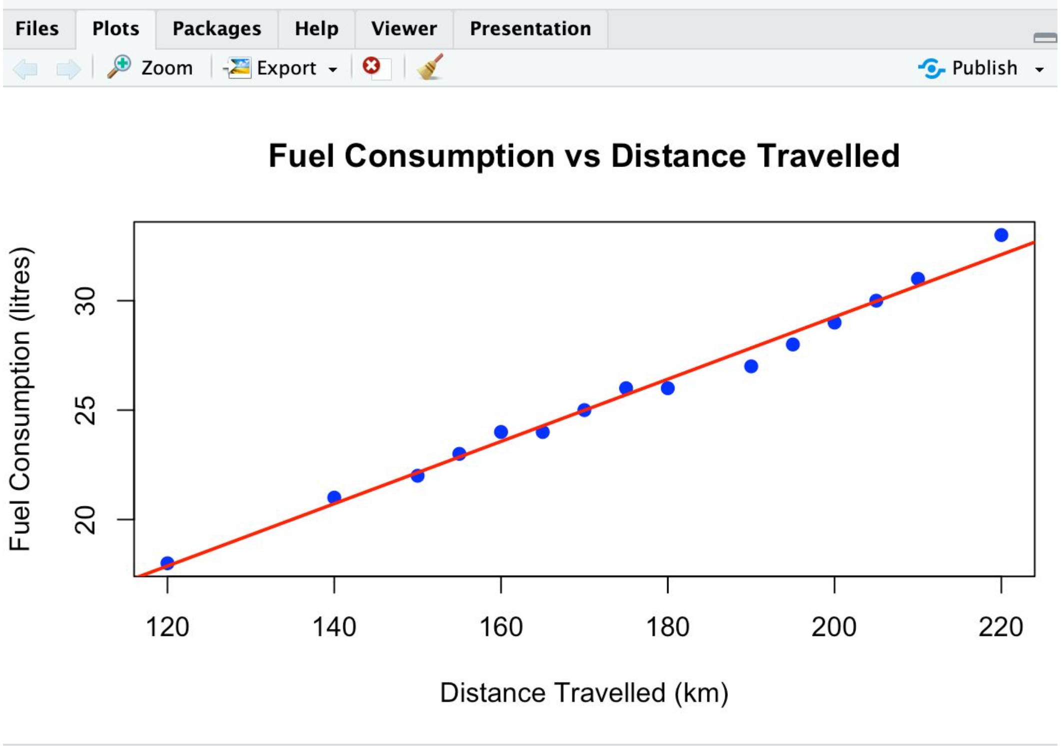
R 4.5.2 · ~/dhcnsdfb/

```
>
> # ------------------------------
> # 3. Regression Equation
> # ------------------------------
>
> intercept <- coef(model)[1]
> slope <- coef(model)[2]
>
> cat("Regression Equation:\n")
Regression Equation:
> cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")
Y = 0.7941 + 0.1423 X

>
> # ------------------------------
> # 4. Prediction for X = 185 km
> # ------------------------------
>
> predicted_value <- predict(model, newdata = data.frame(X = 185))
>
> cat("Estimated fuel consumption for 185 km:",
+     round(predicted_value,4), "litres\n")
Estimated fuel consumption for 185 km: 27.1286 litres
>
> # ------------------------------
> # 5. Scatter Plot with Regression Line
> # ------------------------------
>
> plot(X, Y,
+      main = "Fuel Consumption vs Distance Travelled",
+      xlab = "Distance Travelled (km)",
+      ylab = "Fuel Consumption (litres)",
+      pch = 19,
+      col = "blue")
>
> abline(model, col = "red", lwd = 2)
>
>
```

OUTPUT:-

R Global Environment

Data	
model	List of 12
spearman_test	List of 8
Values	
d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
n	10L
predicted_value	Named num 27.1
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152



Que-3(code part):-

----

1. Enter the Data

Distance Travelled (km)

X <- c(120,150,180,200,160,170,140,210,190,220, 155,165,175,205,195)

Fuel Consumption (litres)

Y <- c(18,22,26,29,24,25,21,31,27,33, 23,24,26,30,28)

----

2. Fit Linear Regression Model

----

model <- lm(Y ~ X)

Display model summary

summary(model)

----

3. Regression Equation

```
intercept <- coef(model)[1] slope <- coef(model)[2]

cat("Regression Equation:\n") cat("Y =", round(intercept,4), "+", round(slope,4), "X\n\n")
```

----

4. Prediction for X = 185 km

```
predicted_value <- predict(model, newdata = data.frame(X = 185))

cat("Estimated fuel consumption for 185 km:", round(predicted_value,4), "litres\n")
```

----

5. Scatter Plot with Regression Line

```
plot(X, Y, main = "Fuel Consumption vs Distance Travelled", xlab = "Distance Travelled
(km)", ylab = "Fuel Consumption (litres)", pch = 19, col = "blue")

abline(model, col = "red", lwd = 2)
```

Question –4

4. A retail company wants to analyse how monthly sales revenue depends on advertising expenditure and store footfall. For 15 randomly selected months, the following data were recorded:

Month	Advertising Expenditure (₹ '000) (X_1)	Average Daily Footfall (X_2)	Sales Revenue (₹ '000) (Y)
1	25	180	410
2	30	200	455
3	28	190	435
4	35	220	500
5	32	210	480
6	27	185	420
7	40	240	550
8	38	230	530
9	34	215	490
10	45	260	590
11	33	205	470
12	29	195	440
13	42	250	570
14	36	225	510
15	31	200	460

Fit a multiple linear regression model of Sales Revenue (Y) on Advertising Expenditure (X_1) and Footfall (X_2). Estimate the expected sales revenue when advertising expenditure is ₹37,000 and footfall is 235.

Code:-

Untitled.R*

```r
# --------------------------------
# 1. Enter the Data
# --------------------------------

# Advertising Expenditure (₹ '000)
X1 <- c(25,30,28,35,32,27,40,38,34,45,
        33,29,42,36,31)

# Average Daily Footfall
X2 <- c(180,200,190,220,210,185,240,230,215,260,
        205,195,250,225,200)

# Sales Revenue (₹ '000)
Y <- c(410,455,435,500,480,420,550,530,490,590,
       470,440,570,510,460)

# --------------------------------
# 2. Fit Multiple Linear Regression
# --------------------------------

model <- lm(Y ~ X1 + X2)

summary(model)

# --------------------------------
# 3. Regression Equation
# --------------------------------

coef(model)

cat("\nRegression Equation:\n")
cat("Y =",
    round(coef(model)[1],4), "+",
    round(coef(model)[2],4),"X1 +",
    round(coef(model)[3],4),"X2\n")

```

Console:-

Console | Terminal | Background Jobs

R 4.5.2 · ~/dhcnsdfb/

```
> # 3. Scatter Plot with Regression Line
> # ------------------------------
> > plot(X, Y,
+      main = "Fuel Consumption vs Distance Travelled",
+      xlab = "Distance Travelled (km)",
+      ylab = "Fuel Consumption (litres)",
+      pch = 19,
+      col = "blue")
> > abline(model, col = "red", lwd = 2)
> > > >
> source("~/dhcnsdfb/Untitled.R")

Regression Equation:
Y = 40.1093 + 2.3094 X1 + 1.7292 X2

Estimated Sales Revenue (₹ '000): 531.9211
>
```

Output:-

Environment | History | Connections | Tutorial

Import Dataset · 112 MiB · List

R · Global Environment

Data	
model	List of 12
new_data	1 obs. of 2 variables
spearman_test	List of 8
Values	
d	num [1:10] -1 1 1 0 1 -1 -1 -1 0 1
d2	num [1:10] 1 1 1 0 1 1 1 1 0 1
intercept	Named num 0.794
n	10L
predicted_sales	Named num 532
predicted_value	Named num 27.1
rank_X	num [1:10] 7 4 10 2 8 3 5 9 1 6
rank_Y	num [1:10] 8 3 9 2 7 4 6 10 1 5
rho_manual	0.951515151515152
slope	Named num 0.142
spearman_corr	0.951515151515151
X	num [1:15] 120 150 180 200 160 170 140 210 190 220 ...
X1	num [1:15] 25 30 28 35 32 27 40 38 34 45 ...
X2	num [1:15] 180 200 190 220 210 185 240 230 215 260 ...
Y	num [1:15] 41 num [1:15] 180 200 190 220 210 185 240 230 215 260 ...

Question-4(code)

----

1. Enter the Data

Advertising Expenditure (₹ '000)

X1 <- c(25,30,28,35,32,27,40,38,34,45, 33,29,42,36,31)

Average Daily Footfall

X2 <- c(180,200,190,220,210,185,240,230,215,260, 205,195,250,225,200)

Sales Revenue (₹ '000)

Y <- c(410,455,435,500,480,420,550,530,490,590, 470,440,570,510,460)

----

2. Fit Multiple Linear Regression

----

model <- lm(Y ~ X1 + X2)

summary(model)

----

3. Regression Equation

----

coef(model)

cat("\nRegression Equation:\n") cat("Y =", round(coef(model)[1],4), "+", round(coef(model)[2],4),"X1 +", round(coef(model)[3],4),"X2\n")

----

4. Prediction

Advertising = 37 (₹37,000)

Footfall = 235

----

new_data <- data.frame(X1 = 37, X2 = 235)

predicted_sales <- predict(model, new_data)

cat("\nEstimated Sales Revenue (₹ '000):", round(predicted_sales,4), "\n")